

1. Unified Big Data Architecture for retail.

→ A retail organization can use a data lake architecture to combine data from websites, mobile apps, physical address and CRM systems.

→ This creates a 360 degree customer view and supports personalized recommendation and targeted marketing.

Proposed Architecture

a) Data Sensors

b) Data Ingestion

c) Processing

d) Analytics and ML

e) Visualization.

2. Distributed Computing for Scientific Research Data.

→ Scientific research often produces very large datasets that cannot be efficiently processed by a single computer.

→ Parallel Processing allows these computers to work simultaneously.

Advantages over Centralized system

- Faster processing
- Better scalability
- Improved resource utilization
- Handles large datasets.

3. Fraud Detection framework for Digital Payments.

A fraud detection framework should process transaction data in real time.

Payment transactions → Streaming Platform → Real-time processing
Alert ← Fraud score ← ML Fraud model

Technologies:-

- Apache Kafka
- Apache Spark Streaming
- machine learning models
- Database.

4. Streaming and processing multimedia data:

Multimedia data such as images, Audio and video creates challenges because of its large size, high processing requirements - different formats.

Challenges:-

- High storage requirements.

- * slow data transfer
- * large processing requirements
- * Different file formats
- * metadata management

Recommendation Technologies

- * cloud object storage

- * HDFS
- * Apache spark
- * Kafka
- * CDN

5. Disaster Recovery and Business Continuity Strategy.

A cloud-based big data platform should maintain continuous operations even when failures occur.

Strategy:-

1. store data in multiple locations.
2. maintain primary and secondary systems
3. use automated systems.

Key mechanisms.

1. Redundancy
2. Replication
3. Automated Failures.

6. Data Quality and Data Governance.

Poor data quality reduces the reliability of analytical results.

Major Issues:-

Inconsistency, Duplication, missing values

Data generation strategies.

1. Data validation rules
2. Data standardization
3. Missing value handling
4. Regular data quality audits.

7. Scalable Recommendation system for streaming services.

A streaming platform can analyze user behavior to recommend relevant content.

User Activity → Kafka → Data Lake → Spark processing → Recommendation models → Personalized Recommendations.

Data Analyzed.

1. Viewing history
2. Search history
3. Watch time
4. Likes and ratings
5. Content preferences.

8. Role of cloud computing in Big Data.

Cloud computing provides infrastructure and services for storing and processing large volume of data.

Advantages:

→ Scalability

2. Flexibility

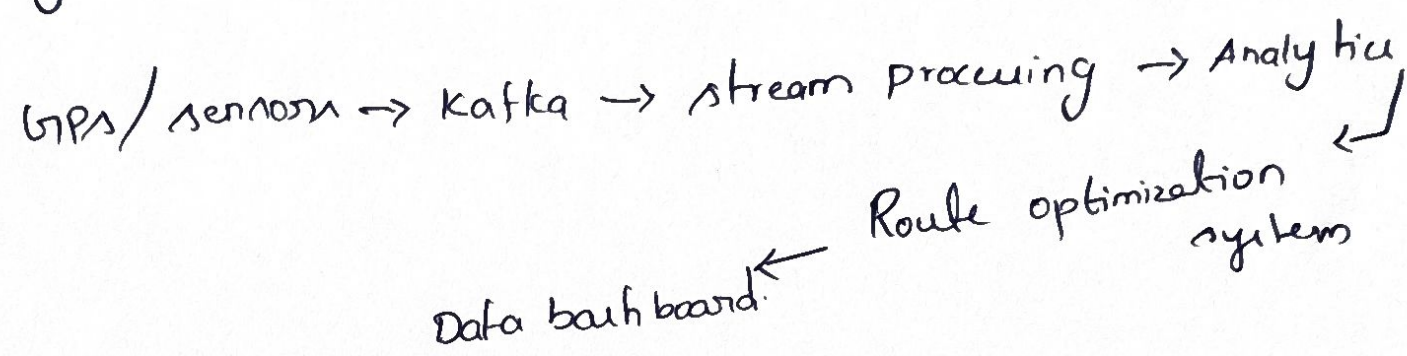
3. cost Efficiency

4. Global access

5. managed services.

9. Real-time Analytics for Transportation and Logistics.

A transportation company can collect real-time data from GPS devices, vehicles, sensors, traffic systems and fuel-monitoring systems.



10. Ethical and legal challenges in big data.

Big data collection creates important ethical and legal concerns

challenge

a) Privacy Violations

b) Lack of informed consent

c) Data breaches

d) Lack of transparency

Recommended policies

- Collect only necessary data
- obtain clear user consent
- use encryption and access control
- perform regular security audits